

Mohammad Ghufuran

520.607.1905 | Tucson, AZ | ghufuran1942@arizona.edu | <http://www.linkedin.com/in/ghufuran1942>

EDUCATION

Master of Engineering in Robotics and Automation, GPA: 3.67

May 2024

University of Arizona

Bachelor of Technology in Mechanical Engineering, GPA: 3.48

May 2022

Aligarh Muslim University, India

TECHNICAL SKILLS

Languages: Python, C++, C, HTML, JavaScript, CSS, MATLAB

Software: SOLIDWORKS, Fusion 360, AutoCAD, DS Abaqus, ANSYS, Adobe Creative Cloud, ROS

Tools & Skills: CAD Modelling, Data Analysis, 3D-Printing, computer networking, Casting, Soldering, Lathes

ACADEMIC EXPERIENCE

University of Arizona

Present

Student Researcher

- Designed and optimized drone design with SOLIDWORKS, ensuring aerodynamics, stability, and functionality while conducting precise modeling and simulation, resulting in a **50% reduction** in size **without performance compromise**
- Deployed the Astar Algorithm in Python for a multi-agent system, resulting in optimized functionality, efficient pathfinding, and refined **performance in various environments**
- Simulated a swarm of 7 drones using Gazebo, enabling an accurate representation of their behavior and performance, validating the system, identifying potential issues, and reducing **hardware adjustments and expenses**

Engineering & Environmental Solutions Pvt Ltd., India

Aug 2022

Drone Development Intern

- Developed and led testing of an agriculture drone integrating IFFCO's nano urea, **reducing required fertilizer by up to 60%**, in collaboration with agriculture experts to ensure usability, resulting in **reduced costs**
- Created DGCA-compliant user and maintenance manual by collaborating with cross-functional teams, which was integral in obtaining product approval, including comprehensive safety, operation, and maintenance information

ACADEMIC PROJECTS

Stem Curriculum Recommender

Present

SIE533(Fundamentals of Data Science for Engineers) Project

- Established a project goal to contribute to **Sustainable Development Goal 4 (UN)**, Quality Education, by developing a **regression model** that matches educational content with specific topics in a curriculum
- Develop a **recommendation system** that will train a machine learning model on open-sourced K-12 educational materials organized by topic and language, empowering educators with guidance and support in selecting and implementing personalized learning resources aligned with their academic goals and learners' needs

Reconfigurable UAV

Present

University of Arizona

- Utilized SOLIDWORKS to design various components, resulting in the **improved structural integrity** of the final model, ultimately contributing to performance, efficiency, and product quality
- Conducted in-depth assessments of various autopilot platforms and successfully implemented the most reliable and high-performing solutions, resulting in **optimized flight control and reliability** of the UAV

Development of Sanitization Drone

Apr 2022

Final Year Project | Best Project Award

- Created an aerodynamic drone with SOLIDWORKS and verified its integrity by 3D printing a design model and simulating it with Ansys to achieve a minimum factor of safety of 10.11
- Validated the simulation results in a wind tunnel, ensuring the drone's stability and optimal performance

RESEARCH PAPER

Experimental Evaluation of Multi-Agent System Affine Transformation Optimization (Under Review)